Spatial Query Branch Ablation

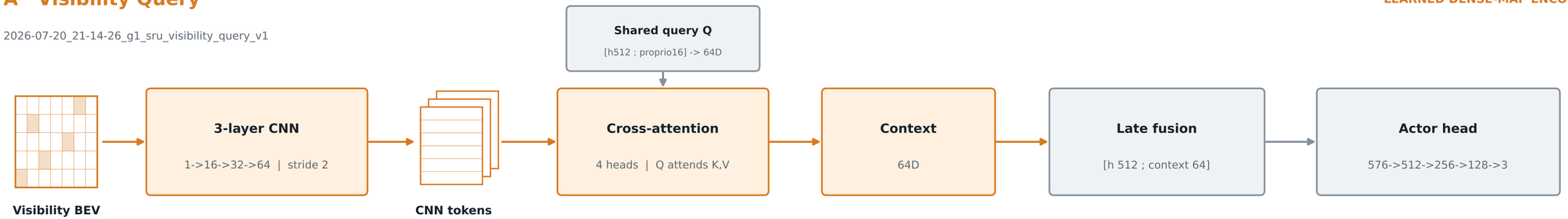
instinct_nav | July 2026

The recurrent backbone and policy head are identical; only the highlighted spatial representation changes.

A Visibility Query

LEARNED DENSE-MAP ENCODING

2026-07-20_21-14-26_g1_sru_visibility_query_v1



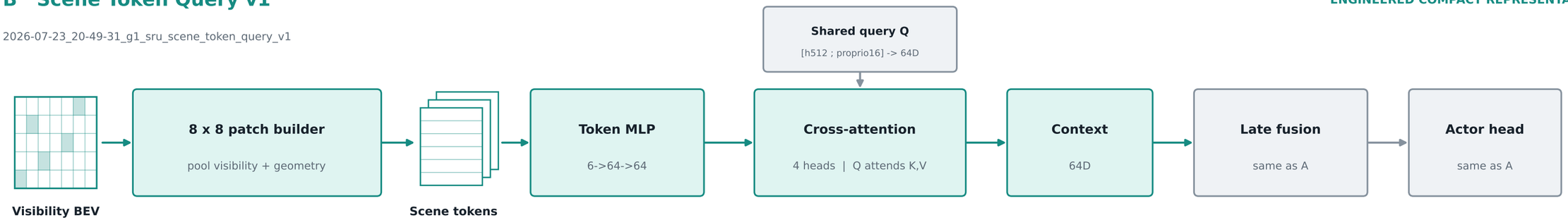
The policy learns spatial features directly from every BEV cell.

branch 99,264 params | actor obs 6672D

B Scene Token Query v1

ENGINEERED COMPACT REPRESENTATION

2026-07-23_20-49-31_g1_sru_scene_token_query_v1



6D/token = (x, y, visibility, goal distance, goal cos, goal sin)

branch 80,320 params | actor obs 2960D

KEY DIFFERENCE dense learned CNN features (A) vs. compact, explicitly structured 6D tokens (B)

Not shown: identical depth encoder, Actor SRU (80->512), critic, and frozen G1 low-level controller.

Both branches are queried after the SRU; neither is written into recurrent memory.